Week 44: The bag that copies

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

Problems and routes

Problem	K–1	2–3	4–5	route
Bags after two draws	P1			first
Bags after three draws	P2			first
Every story to three red and three blue	P3			return
One draw back	P4			return
Two-draw color stories		P1		first
Every marked two-draw story		P2		first
Copying against return only		P3		return
Stories to three red and two blue		P4		return
Stories for four-draw bags		P5		return
Which four-draw bag is more likely?		P6		return
Two-draw marked stories			P1	first
Return without a copy			P2	first
Does order change the chance?			P3	first
Twenty-four marked histories			P4	return
Four draws without a 120-row list			P5	return
Five draws and more			P6	return

First route. K–1 P1–P2; Grades 2–3 P1–P2; Grades 4–5 P1–P3. **Return visits.** K–1 P3–P4; Grades 2–3 P3–P6; Grades 4–5 P4–P6.

Key, one entry per problem

Bags after two draws

K–1 P1 · first route

Idea. After two draws the bag is 3R/1B, 2R/2B or 1R/3B; only the middle one has two color stories (RB and BR).

After two draws the possible bags are **3R/1B** from RR, **2R/2B** from RB or BR, and **1R/3B** from BB. Only the middle bag has two different two-draw *color* stories. A child who counts marked counters instead finds two stories for every bag (R1 R1 or R1 R2; R1 B1 or B1 R1; B1 B1 or B1 B2): accept that, then ask which bags have two different color orders. This is a possibility question; counting these four words as equally likely would be wrong for random copying.

Bags after three draws

K–1 P2 · first route

Idea. Every three-draw bag keeps at least one counter of each color, so five red and no blue is impossible.

For the printed choices: 4R/1B is possible via RRR; 2R/3B via RBB, BRB, or BBR; 5R/0B is impossible; 1R/4B via BBB. The full collection of possible three-draw bags also includes 3R/2B, via RRB, RBR, or BRR: ask for it as a follow-up. The all-red picture is impossible because the original blue is never removed.

Every story to three red and three blue

K–1 P3 · return route

Idea. Three red and three blue after four draws needs two red and two blue draws: six color stories, one for each choice of two red positions.

The displayed 3R/3B finish requires exactly two red draws and two blue draws. All six color stories are **RRBB, RBRB, RBBR, BRRB, BRBR, BBRR**. The page has eight story rows, two more than needed. Completeness can be seen by choosing the two red positions among four: (1,2), (1,3), (1,4), (2,3), (2,4), (3,4).

The four-draw bags with more blue than red are exactly **1R/5B**, made by BBBB, and **2R/4B**, made by RBBB, BRBB, BBRB, or BBBR. The bag has six counters, with at least one of each color; the other mixtures are tied or red-majority. The two bag boxes, each with a story row, hold these. Children need a story for each blue-majority bag, not necessarily all four stories for the second bag.

One draw back

K–1 P4 · return route

Idea. Removing one possible last copy gives each predecessor; a predecessor keeps both colors, so copying never loses a color.

5R / 1B: 4R / 1B, R.

4R / 2B: 3R / 2B; 4R / 1B, R; B.

3R / 3B: 2R / 3B; 3R / 2B, R; B.

Each last draw adds exactly one counter of its own color, so remove one candidate last copy to find a predecessor. A predecessor must still contain at least one red and one blue. For example, removing the only blue from 5R/1B would make 5R/0B, which cannot have arisen from the required start.

Copying cannot lose all blue counters, or all red counters. Every drawn counter is returned, and each move only adds a counter. In particular, the original blue remains. This is a deterministic invariant, not an observation about which color seems to be winning.

Two-draw color stories

Grades 2–3 P1 · first route

Idea. RR, RB, BR, BB lead to 3R/1B, 2R/2B, 2R/2B, 1R/3B; color-story counts alone do not measure chance.

RR $\to$ 3R/1B; RB $\to$ 2R/2B; BR $\to$ 2R/2B; BB $\to$ 1R/3B. The balanced finish has two color stories; each extreme has one. Color-story counts alone do not measure probability here.

Every marked two-draw story

Grades 2–3 P2 · first route

Idea. The six marked histories are equally likely, two for each final bag, so all three bags have chance $1/3$.

All six marked histories are listed below. Each first choice has chance 1/2; each second choice, conditional on the first, has chance 1/3. Thus every row has chance 1/6.

R1: R1, 3R / 1B, 2.

R1: R2, 3R / 1B, 2.

R1: B1, 2R / 2B, 1.

B1: R1, 2R / 2B, 1.

B1: B1, 1R / 3B, 0.

B1: B2, 1R / 3B, 0.

All three compositions have two histories, so none has the most: each has probability 1/3. Notice that RR and BB each merge two marked histories, while RB and BR each merge one.

Copying against return only

Grades 2–3 P3 · return route

Idea. Copying ties zero, one and two red draws at $1/3$ each; return only gives $1/4$, $1/2$, $1/4$.

Copying: zero, one, two red draws are tied at 2/6 = 1/3 each. **Return only:** RR, RB, BR, BB are now four equally likely histories, so the totals have weights 1, 2, 1 and probabilities 1/4, 1/2, 1/4. One red draw is most likely in the unchanged bag. Copying increases the chance of repeating the first color.

Stories to three red and two blue

Grades 2–3 P4 · return route

Idea. RRB, RBR and BRR each have chance $1/12$: reordering a color story does not change its chance.

The stories are RRB, RBR, BRR. Each has chance 1/12: RRB gives (1/2)(2/3)(1/4); RBR gives (1/2)(1/3)(2/4); BRR gives (1/2)(1/3)(2/4). Each represents two of the 24 marked histories.

Stories for four-draw bags

Grades 2–3 P5 · return route

Idea. Five red and one blue has one four-draw story; four red and two blue has four; three red and three blue has six, one for each choice of the two red draws.

5R/1B: only RRRR, because all four added counters must be red. **4R/2B:** four stories, RRRB, RRBR, RBRR, BRRR. **3R/3B:** six stories, RRBB, RBRB, RBBR, BRRB, BRBR, BBRR, one for each choice of the two red draws.

Which four-draw bag is more likely?

Grades 2–3 P6 · return route

Idea. After four copying draws, five red and one blue and three red and three blue each have chance $1/5$: RRRR has 24 of the 120 marked histories, and each of the six 3R/3B color stories has 4.

Neither: each has chance 1/5. RRRR has $1 \times 2 \times 3 \times 4 = 24$ of the 120 marked histories, since its red draws have 1, 2, 3 and 4 red counters to choose from. Each of the six 3R/3B color stories has $1 \times 2 \times 1 \times 2 = 4$, so that bag also has 24. An adult can carry the multiplication.

Two-draw marked stories

Grades 4–5 P1 · first route

Idea. The six marked histories give two each to zero, one and two red draws: all totals tie at $1/3$.

The six marked histories are (R1,R1), (R1,R2), (R1,B1), (B1,R1), (B1,B1), (B1,B2). Each has probability 1/6. The first two have two red draws, the middle two have one, and the last two have zero. All totals are tied, two histories each.

Return without a copy

Grades 4–5 P2 · first route

Idea. Without copies the totals have chances $1/4$, $1/2$, $1/4$ instead of $1/3$ each.

Without copies, the four histories are RR, RB, BR, BB, each with probability 1/4. The probabilities for zero, one, two reds become 1/4, 1/2, 1/4, compared with 1/3, 1/3, 1/3 under copying. In copying, the first color gains an extra identity for the second draw, giving RR and BB two marked histories each.

Does order change the chance?

Grades 4–5 P3 · first route

Idea. RRB, RBR and BRR each have chance $1/12$: the same factors in a different order, though the draws are dependent.

RRB, RBR, BRR each has probability 1/12. Their products are (1/2)(2/3)(1/4), (1/2)(1/3)(2/4), and (1/2)(1/3)(2/4). Reordering does not change the product of the red factors and blue factors. The draws remain dependent despite this equality.

Twenty-four marked histories

Grades 4–5 P4 · return route

Idea. After three draws each red total 0, 1, 2, 3 has six of the 24 marked histories.

There are six marked histories for each total 0, 1, 2, 3. BBB and RRR each have 1 $\times$ 2 $\times$ 3 = 6 identity histories. For one red, the three words RBB, BRB, BBR have two histories each, hence six. The three words with two reds also have two each. These four groups cover all eight possible color words and all 24 marked histories.

Four draws without a 120-row list

Grades 4–5 P5 · return route

Idea. Extending each three-draw group by the five fourth-draw identities gives 24 of 120 histories to every total 0 to 4: chance $1/5$ each.

Each of the five totals has 24 of the 120 marked histories, hence chance 1/5. Extend the six histories in each three-draw group by the five available fourth-draw identities. The contributions to totals 0 through 4 are:

0: 6 $\times$ 4, 0, 24.

1: 6 $\times$ 3, 6 $\times$ 1, 24.

2: 6 $\times$ 2, 6 $\times$ 2, 24.

3: 6 $\times$ 1, 6 $\times$ 3, 24.

4: 0, 6 $\times$ 4, 24.

Five draws and more

Grades 4–5 P6 · return route

Idea. After n draws each red total from 0 to n has chance $1/(n+1)$.

Predict 1/6 for each red total after five draws, and 1/(n+1) after n draws. Proof: a word with r reds has probability r!(n-r)!/(n+1)!, since its red numerators are 1,...,r and its blue numerators 1,...,n-r. There are n!/[r!(n-r)!] such words. Multiplication gives 1/(n+1). A child's observed pattern remains a conjecture until justified; the general proof is an optional endpoint invited by the problem.